

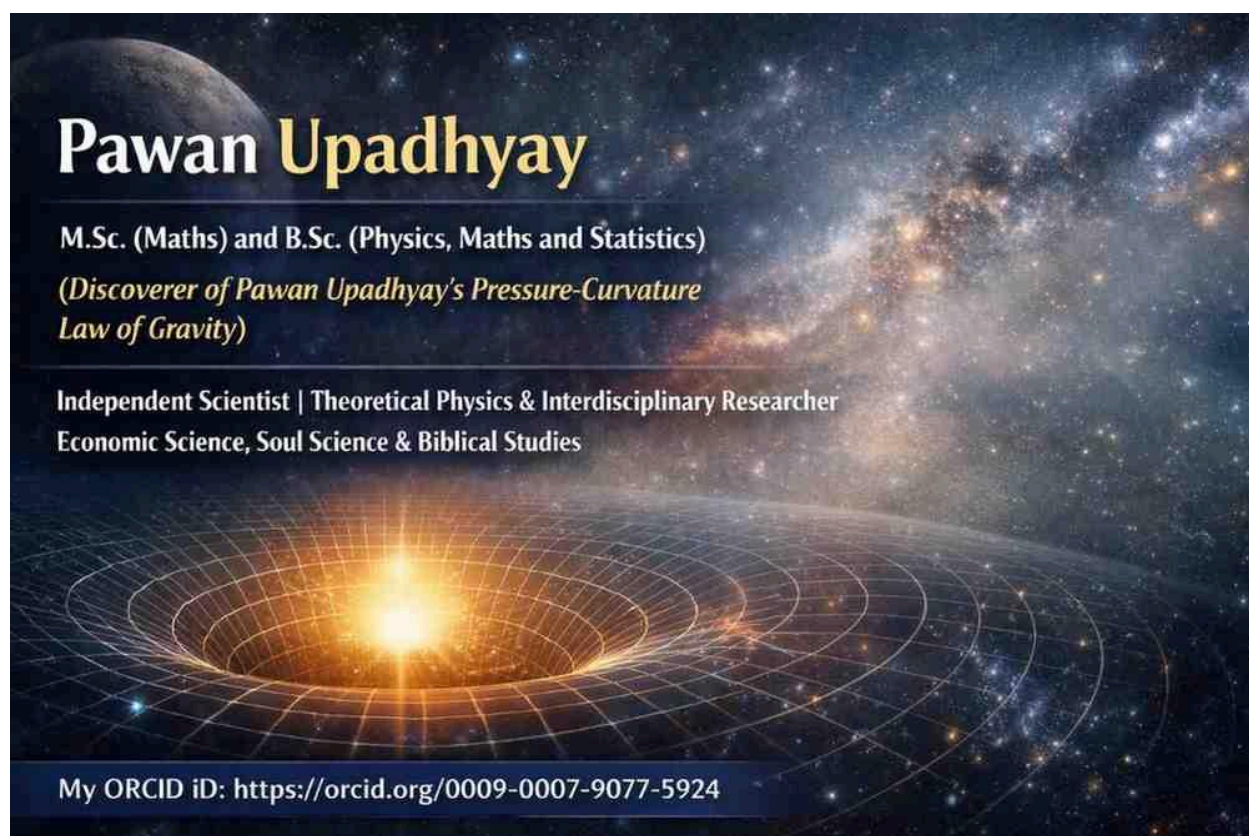
Gravitational Field and Potential in the PPC Law of Gravity

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Abstract

This paper formulates the gravitational field within Pawan Upadhyay's Pressure Curvature (PPC) Law of Gravity. We establish a direct correspondence between the classical potential formulation of gravity and the PPC pressure-based description. By identifying gravitational potential with pressure per unit density, we derive consistent expressions for the gravitational

field, circular motion, and weak-field limits. The framework unifies force density, pressure gradients, and potential theory into a single description.

1. Introduction

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In classical physics, gravity is described using gravitational potential ϕ , where the gravitational field is given by:

$$\mathbf{g} = -\nabla\phi$$

In the PPC framework, gravity is instead interpreted as a pressure-driven phenomenon:

$$F = \rho a = -\nabla P_g$$

This paper connects these two descriptions and shows their equivalence under appropriate identification.

2. PPC Fundamental Equation

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The fundamental PPC relation is:

$$\rho \frac{dv}{dt} = -\nabla P_g$$

Dividing by ρ :

$$a = -\frac{1}{\rho} \nabla P_g$$

This expresses acceleration in terms of the gradient of gravitational pressure.

3. Gravitational Field Definition

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In classical form:

$$\mathbf{g} = -\nabla\phi$$

Comparing with PPC:

$$\mathbf{a} = -\frac{1}{\rho}\nabla P_g$$

we identify:

$$\mathbf{g} = \mathbf{a}$$

Thus:

$$-\nabla\phi = -\frac{1}{\rho}\nabla P_g$$

which implies:

$$\phi = \frac{P_g}{\rho}$$

4. Interpretation of Potential

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This result provides a new physical meaning of gravitational potential:

- Potential is not an abstract scalar
- It is directly related to **pressure per unit density**

Thus:

$$\phi = \frac{P_g}{\rho}$$

serves as the bridge between classical and PPC gravity.

5. Weak-Field Limit

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For weak gravitational fields, classical results must be recovered.

For a point mass M :

$$\phi = -\frac{GM}{r}$$

Using PPC relation:

$$P_g = \rho\phi = -\frac{\rho GM}{r}$$

Taking gradient:

$$-\frac{1}{\rho}\nabla P_g = \frac{GM}{r^2}$$

Thus:

$$a = \frac{GM}{r^2}$$

which matches Newtonian gravity.

6. Circular Motion

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For circular motion:

$$a = \frac{v^2}{r}$$

Using PPC:

$$\rho \frac{v^2}{r} = -\nabla P_g$$

Substituting the pressure field derived above yields:

$$v^2 = \frac{GM}{r}$$

7. Consistency with Kepler's Laws

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From $v^2 = GM/r$, we obtain:

$$T^2 = \frac{4\pi^2}{GM} r^3$$

Thus, Kepler's third law is recovered.

Since the force is radial:

- Angular momentum is conserved
- Equal areas are swept in equal times

Hence, Kepler's second law follows.

The inverse-square nature ensures elliptical orbits, satisfying Kepler's first law.

8. Role of Pressure in Curvature

In PPC theory:

Mass Density $\rightarrow$ Energy Density $\rightarrow$ Pressure $\rightarrow$ Stress–Energy Tensor $\rightarrow$ Curvature

Thus:

- Pressure determines stress–energy
- Stress–energy determines metric
- Metric determines geodesics

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9. Unified Expression

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Combining all results:

$$\mathbf{g} = -\nabla\phi = -\frac{1}{\rho}\nabla P_g$$

This expresses gravitational field in both classical and PPC forms.

10. Conclusion

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The PPC framework provides a consistent reinterpretation of gravitational potential and field. By identifying potential as pressure per unit density, the gravitational field emerges naturally from pressure gradients. This formulation reproduces Newtonian gravity and Keplerian motion in the weak-field limit while offering a deeper physical mechanism based on pressure.

Keywords

PPC Gravity, Gravitational Field, Potential, Pressure Gradient, Force Density, Weak Field Limit, Kepler's Laws

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Gravitational Field Definition

$$\mathbf{g} = -\nabla\phi$$

Meaning

- $\mathbf{g} \rightarrow$ gravitational field (acceleration per unit mass)
- $\phi \rightarrow$ gravitational potential
- $\nabla\phi \rightarrow$ spatial rate of change of potential

👉 The negative sign means:

- The field points **toward decreasing potential**
- Objects move **“downhill” in potential**

Physical Interpretation

- If potential changes in space $\rightarrow$ a gravitational field exists
- If potential is constant $\rightarrow$ no gravity

👉 So:

Gravity is the gradient (slope) of potential

If:

$$\phi = \text{constant}$$

Then:

$$\nabla \phi = 0$$

So:

$$\mathbf{g} = 0$$

$$\mathbf{g} = -\frac{1}{\rho} \nabla P_g$$

If potential is constant, then:

$$P_g = \rho\phi = \text{constant}$$

So:

$$\nabla P_g = 0 \Rightarrow \mathbf{g} = 0$$

5. Unified statement

 **No gradient $\rightarrow$ no gravity**

This applies to:

- Potential field
- Pressure field
- Force field

 **Constant potential $\Rightarrow$ zero gradient $\Rightarrow$ zero gravity.**

Key Identification

$$\phi = \frac{P_g}{\rho}$$

👉 This is a central PPC relation:

- Potential = **pressure per unit density**
- Gravitational field = **pressure gradient per density**

Unified View

$$\mathbf{g} = -\nabla\phi = -\frac{1}{\rho}\nabla P_g$$

Final Insight

👉 *In PPC, the gravitational field is simply the normalized gradient of gravitational pressure.*

For a spherically symmetric function:

$$\nabla\phi = \frac{d\phi}{dr} \hat{\mathbf{r}}$$

Unified Relation: Field $\leftrightarrow$ Pressure $\leftrightarrow$ Gradient

$$\mathbf{g} = -\nabla\phi$$

$$\mathbf{g} = -\frac{1}{\rho}\nabla P_g$$

-
- **Gravitational field** $\mathbf{g}$: acceleration per unit mass (m/s^2)
 - **Gravitational pressure field** P_g : scalar pressure (N/m^2)
 - **Pressure gradient field** ∇P_g : force density (N/m^3)

Check Units of Relation

$$\phi = \frac{P_g}{\rho}$$

1. Units of each quantity

- Gravitational potential $\phi \rightarrow \mathbf{m^2/s^2}$
- Pressure $P_g \rightarrow \mathbf{N/m^2 = kg/(m \cdot s^2)}$
- Density $\rho \rightarrow \mathbf{kg/m^3}$

2. Compute units of P_g/ρ

$$\frac{P_g}{\rho} = \frac{kg/(m \cdot s^2)}{kg/m^3}$$

Invert denominator:

$$= \frac{kg}{m \cdot s^2} \times \frac{m^3}{kg}$$

Cancel kg :

$$= \frac{m^3}{m \cdot s^2} = \frac{m^2}{s^2}$$

3. Final result

$$\frac{P_g}{\rho} = m^2/s^2$$

4. Conclusion

✓ Units match perfectly:

- LHS $\phi \rightarrow m^2/s^2$
- RHS $P_g/\rho \rightarrow m^2/s^2$